

HOSPITAL COURSE SUMMARY

SURGICAL / NEUROSURGICAL INTENSIVE CARE UNIT

Admission: 10/18/2025

Transfer / Disposition: 12/22/2025 — Inpatient Neurologic Rehabilitation (Alder Creek Neuro-Rehabilitation Institute)

Attending of Record (SICU / ongoing care): Andrew Steinbach, MD — Attending Intensivist, Surgical / Neurosurgical Critical Care

Trauma Surgery Attending of Record (admission): Priya Ramaswamy, MD

Principal Diagnosis: Severe traumatic brain injury — diffuse axonal injury

Secondary Diagnoses: Open pelvic ring disruption; bilateral open comminuted tibia/fibula fractures; Grade V splenic laceration status post splenectomy; intraperitoneal bladder-dome laceration status post primary repair; multiple right rib fractures with flail segment, right hemopneumothorax, and bilateral pulmonary contusions; hemorrhagic shock with acute traumatic coagulopathy; ventilator-associated pneumonia (treated); catheter-associated urinary tract infection (treated); ICU-acquired weakness; protein-calorie malnutrition; post-traumatic amnesia.

SUMMARY

Ms. Yolanda Reyes is a 34-year-old woman admitted on 10/18/2025 following a high-energy auto-versus-pedestrian collision, in which she was struck at approximately 30 mph by a transit bus while standing at a curbside bus stop. She sustained catastrophic polytrauma dominated by a severe closed-head injury (diffuse axonal injury), a vertically unstable open pelvic ring disruption, bilateral open tibia/fibula fractures, and intra-abdominal hemorrhage from a shattered spleen with an associated bladder-dome laceration. She required emergent damage-control laparotomy and combined-case external skeletal stabilization on the day of admission, followed by a staged reconstructive orthopedic course culminating in definitive open reduction and internal fixation of the pelvic ring and both tibiae on 10/25/2025, and by tracheostomy with concurrent percutaneous endoscopic gastrostomy on 11/05/2025.

Throughout the sixty-five-day surgical intensive care course summarized below, Ms. Reyes was persistently non-decisional and unable to communicate meaningful preferences regarding her medical care. She was maintained in a medically induced coma with continuous deep sedation for cerebral protection during approximately the first three weeks of admission; she then emerged into a prolonged, agitated state of post-traumatic amnesia through the middle and end of November; and only in the second and third weeks of December did she demonstrate the earliest, still-fragmentary signs of functional cognitive recovery. Surrogate decision-making was carried by her sister, Ms. Marisela Reyes, from admission forward, with a temporary probate conservatorship granted on 11/10/2025 formalizing that authority for the balance of the acute admission. At the point of transfer she remained dependent for all activities of daily living, required continuous supervision, and was intermittently oriented to self only.

She is transferred today, 12/22/2025, in stable condition to inpatient neurologic rehabilitation for continued weaning, mobilization, and cognitive rehabilitation.

COURSE BY SYSTEM

Neurologic

Initial ground-EMS Glasgow Coma Scale was 5 (E1 V1 M3) with agonal respirations; field rapid-sequence intubation was performed. Admission head CT demonstrated **diffuse axonal injury** with scattered petechial hemorrhages at the gray-white matter junctions bilaterally, in the splenium of the corpus callosum, and in the dorsal midbrain, together with early diffuse cerebral edema and effacement of the cortical sulci. There was no operative extra-axial hematoma requiring evacuation; in view of the diffuse edema, an emergent decompressive craniectomy was performed together with placement of an intraparenchymal intracranial-pressure monitor by neurosurgery on 10/18/2025.

For the first three weeks of admission (10/18/2025 through approximately 11/07/2025), Ms. Reyes was maintained in a medically **induced coma** with continuous propofol and fentanyl infusions, titrated to a Richmond Agitation-Sedation Scale (RASS) of -4 to -5, with cisatracurium neuromuscular blockade during the first seventy-two hours to control

intracranial pressure and to permit lung-protective ventilation in the setting of bilateral pulmonary contusions and flail chest. ICP was maintained below 20 mmHg with head-of-bed elevation, normothermia, brief hyperosmolar therapy with 3% saline (10/18 to 10/22), and one course of mannitol on 10/19. Seizure prophylaxis with levetiracetam was continued throughout the acute phase. The ICP monitor was removed on 10/28/2025 after seventy-two hours of consistently normal readings off osmotherapy.

Sedation was gradually weaned beginning 11/08/2025 following successful tracheostomy placement. Emergence was slow and non-linear. Between mid-November and the end of November, Ms. Reyes displayed the classic constellation of **post-traumatic amnesia**: fluctuating arousal with eye-opening to voice but inconsistent visual tracking; profound disorientation to time, place, and situation; marked motor restlessness and agitation requiring scheduled quetiapine and as-needed dexmedetomidine boluses; inability to form or retrieve new memories from day to day; and no reliable ability to follow multi-step commands. The Galveston Orientation and Amnesia Test (GOAT) was administered at bedside on 11/18, 11/22, and 11/29 with scores of 22, 28, and 41 respectively — each well below the cutoff of 75 that defines emergence from post-traumatic amnesia. She did not provide any reliable history, did not participate meaningfully in goals-of-care conversations, and was not asked to make or communicate medical decisions during this interval; all such decisions were made by her surrogate.

Cognitive improvement into December was slow but one-directional. By the second week of December she was intermittently able to state her first name and to recognize her sister at bedside, though she remained disoriented to date, month, and hospital. GOAT scores on 12/08, 12/15, and 12/20 were 54, 63, and 71 respectively — still short of the emergence cutoff at the time of transfer. She could follow simple one-step midline commands (“squeeze my hand,” “open your eyes”) with prompting, but not consistent multi-step commands. Attention span was measured in seconds; short-term recall was essentially absent; executive function could not be assessed. Discharge GCS on 12/22/2025 was 13 (E4 V4 M5) with the verbal component limited by dysarthria and confusion, not by aphasia.

Serial neurosurgical follow-up imaging (MRI 11/12/2025 and repeat CT 12/10/2025) confirmed the pattern of diffuse axonal injury with expected evolution and without any new hemorrhage or hydrocephalus. Neurology and neurosurgery signed off on 12/18/2025 with recommendations for outpatient follow-up.

Neurologic status at transfer: Persistent severe cognitive impairment consistent with resolving post-traumatic amnesia in the setting of moderate-to-severe traumatic brain injury; dependent for all higher-order decision-making; requires 24-hour supervision; not decisional. Transfer to inpatient neurologic rehabilitation.

Pulmonary

Admission chest imaging demonstrated a right-sided flail segment (ribs 4–9), right hemopneumothorax, and bilateral pulmonary contusions worse on the right. A 32-French right-sided chest tube was placed in the trauma bay with return of 350 mL of blood and immediate air leak; the right lung re-expanded on repeat chest radiograph. The right chest tube was removed on 10/24/2025 after resolution of the air leak and complete re-expansion.

Ms. Reyes was maintained on assist-control mechanical ventilation with lung-protective settings (tidal volume 6 mL/kg predicted body weight, plateau pressure < 30 cmH₂O, permissive hypercapnia as tolerated by ICP) throughout the first two-plus weeks of admission. She developed ventilator-associated pneumonia on 10/27/2025, culture-positive for methicillin-sensitive *Staphylococcus aureus* on tracheal aspirate; a seven-day course of intravenous cefazolin was completed on 11/03/2025 with clinical and radiographic resolution.

After nineteen days of translaryngeal intubation, and following failed spontaneous breathing trials on 11/01 and 11/03 attributable to poor cough, copious secretions, and unreliable airway protection given her depressed level of consciousness, an open surgical **tracheostomy** (size 8 Shiley cuffed nonfenestrated tube) was performed on **11/05/2025** in the same anesthetic as PEG placement. Ventilator weaning by pressure-support trials proceeded incrementally through mid-November. She was transitioned to tracheostomy-collar trials in daily lengthening increments beginning 11/17/2025 and was fully off positive-pressure ventilation by 11/22/2025. Speaking-valve trials were tolerated for short intervals from 11/24/2025. The tracheostomy tube was downsized on 11/26/2025 and **decannulated on 11/28/2025** after passage of a capping trial. The stoma was allowed to close by secondary intention and had granulated closed at the time of transfer.

RIVERGATE REGIONAL TRAUMA CENTER

Reyes, Yolanda

Level I Trauma Center

MRN: RRTC-0074318 DOB: 06/22/1991 (34 yrs) Sex: F

2100 Medical Center Drive, Rivergate, CA 91766Adm: 10/18/2025 Xfer: 12/22/2025 Coverage: Medi-Cal / BlueShield of CA HMO

At transfer she is breathing spontaneously on room air, saturating 97–99%, with a productive cough, no supplemental oxygen requirement, and a healing tracheostomy stoma covered with a dry dressing.

Cardiovascular

Ms. Reyes arrived in hemorrhagic shock (heart rate 128, blood pressure 88/54, base deficit –11, lactate 6.4). Massive transfusion protocol was activated at 02:52 on 10/18/2025 with 1:1:1 component resuscitation. Norepinephrine was initiated intraoperatively and continued at low dose for approximately forty-eight hours post-operatively; she was weaned off all vasopressors by 10/21/2025. There was no evidence of blunt cardiac injury on serial troponins and 12-lead ECGs; echocardiography on 10/22/2025 demonstrated preserved biventricular function without pericardial effusion or wall-motion abnormality.

Sinus tachycardia (110–125 bpm) persisted through much of the acute phase, attributed to pain, agitation, autonomic storming secondary to the diffuse axonal injury, and volume shifts; this was managed with scheduled propranolol from 11/10/2025, with good rate control by the third week of November. She experienced no arrhythmias requiring intervention. Blood pressure has been normal off all agents since 10/21/2025.

Venous thromboembolism prophylaxis was managed cautiously in the setting of intracranial injury and multiple bleeding sites: mechanical sequential compression devices from admission; low-dose subcutaneous enoxaparin (30 mg every 12 hours) initiated 10/24/2025 after neurosurgical clearance and stability of the abdominal and pelvic bleeding sources; escalated to prophylactic-dose enoxaparin 40 mg daily from 10/30/2025. Bilateral lower-extremity duplex ultrasound on 11/06 and 12/01 was negative for deep vein thrombosis. No clinical or radiographic pulmonary embolism.

Gastrointestinal / Nutrition

On 10/18/2025 she underwent emergent exploratory laparotomy with total splenectomy (AAST Grade V shattered spleen with hilar injury; approximately 2.5 L hemoperitoneum evacuated), primary two-layer repair of a 2 cm intraperitoneal bladder-dome laceration, four-quadrant abdominal packing, and damage-control temporary abdominal closure with an ABTHERA negative-pressure dressing. She returned to the operating room on 10/20/2025 for planned second-look laparotomy, pack removal, and definitive primary fascial closure, which was tolerated well.

Post-splenectomy vaccinations against *Streptococcus pneumoniae*, *Haemophilus influenzae* type b, and *Neisseria meningitidis* were administered on 10/28/2025 (14 days after splenectomy) per protocol. She will require lifelong asplenia precautions and follow-up boosters.

Enteral nutrition was initiated via nasogastric tube on 10/22/2025 following abdominal closure, at trophic rates initially and advanced to goal over the following forty-eight hours. She tolerated tube feeds throughout without evidence of aspiration, bowel obstruction, or refeeding phenomenon. Durable enteral access was established with a 20-French percutaneous endoscopic gastrostomy (PEG) tube placed by pull technique on 11/05/2025 in the same anesthetic as the tracheostomy. PEG feeds have continued as the primary route of nutrition throughout the balance of the admission; a modified-barium-swallow study on 12/16/2025 demonstrated silent aspiration on thin liquids and she was maintained NPO by mouth at transfer, with all nutrition, hydration, and medications delivered via PEG.

Bowel function was managed with a scheduled regimen of docusate, senna, and as-needed polyethylene glycol; she has had no episodes of *Clostridioides difficile* colitis and no evidence of stress ulceration on scheduled proton-pump inhibitor prophylaxis.

Genitourinary

The 2 cm intraperitoneal bladder-dome laceration identified on admission cystography and confirmed at laparotomy was closed primarily in two layers on 10/18/2025, with a suprapubic Jackson-Pratt drain placed adjacent to the repair. A follow-up cystogram on 10/28/2025 showed a watertight repair without extravasation, and the indwelling Foley catheter was removed on 10/31/2025. She developed a catheter-associated urinary tract infection (*Escherichia coli*, pan-susceptible) diagnosed on 11/02/2025 and was treated with a seven-day course of ceftriaxone. Renal function remained normal throughout with baseline serum creatinine of 0.7–0.9 mg/dL. At transfer she is voiding via intermittent straight-catheterization every six hours, with post-void residuals in the 60–120 mL range, pending further neurogenic-bladder work-up in rehabilitation.

Infectious Disease

Course was marked by two significant hospital-acquired infections, both fully treated: (1) ventilator-associated pneumonia with methicillin-sensitive *Staphylococcus aureus* on 10/27/2025, treated with a seven-day course of intravenous cefazolin; and (2) catheter-associated urinary tract infection with pan-susceptible *Escherichia coli* on 11/02/2025, treated with a seven-day course of ceftriaxone. Serial deep-wound cultures obtained from all three open-fracture sites during the staged washouts (10/18, 10/22, and 10/25) demonstrated no growth. Post-operative surveillance of the pelvic and tibial ORIF wounds through the balance of the admission showed no evidence of surgical-site infection. Blood cultures were sterile on all draws.

Antibiotic prophylaxis included cefazolin plus gentamicin per open-fracture protocol on 10/18/2025 for 72 hours, plus local vancomycin/tobramycin PMMA antibiotic beads at the Gustilo IIIB left ankle site (placed 10/18/2025 and exchanged 10/22/2025; explanted at definitive fixation on 10/25/2025). Infectious disease was consulted for treatment planning of the two hospital-acquired infections and signed off after completion of therapy.

Musculoskeletal / Orthopedic

The orthopedic course was staged, as described in detail in the operative report compilation of record:

- **10/18/2025 (combined-case with damage-control laparotomy):** Application of anterior pelvic external fixator (two supra-acetabular pins per side); bilateral lower-extremity spanning external fixators (knee-to-hindfoot) bridging the tibial fractures; pulsed-lavage irrigation and debridement of the left iliac-crest, right mid-tibial, and left anteromedial ankle open wounds; placement of vancomycin/tobramycin PMMA antibiotic beads at the Gustilo IIIB left ankle site; and application of negative-pressure wound therapy dressings at all three open-fracture sites.
- **10/22/2025:** Second staged washout of all three open-fracture wounds with repeat sharp debridement, deep cultures, exchange of PMMA antibiotic beads at the left ankle, and Wound VAC exchange.
- **10/25/2025:** Definitive open reduction and internal fixation of the left pelvic ring (percutaneous iliosacral screw fixation of the left sacroiliac joint diastasis; anterior symphyseal plating with 6-hole reconstruction plate); reamed intramedullary interlocking nail fixation of the right mid-shaft tibia; ORIF of the left distal tibial pilon fracture (medial locking plate with independent lag-screw fixation of articular fragments) and left distal fibula (one-third tubular plate); removal of all external fixators; and left ankle soft-tissue coverage by Plastic Surgery with a pedicled medial gastrocnemius rotational flap and overlying split-thickness skin graft harvested from the ipsilateral anterolateral thigh.

The postoperative course was uncomplicated from an orthopedic standpoint. The left ankle rotational flap remained well-perfused throughout with 100% graft take documented at the 12/03/2025 combined orthopedic/plastic surgery follow-up. Suprapubic and Blake drains were removed by 11/01/2025. Serial follow-up radiographs of the pelvis and both tibiae have demonstrated maintained hardware position and early callus at the tibial fracture sites without signs of failure or malunion.

Weight-bearing status at transfer: **non-weight-bearing bilateral lower extremities** with anticipated advancement to touch-down weight-bearing on the right at approximately eight weeks post-fixation, and continued non-weight-bearing on the left pending clinical and radiographic milestones (anticipated twelve-plus weeks). Bilateral custom knee immobilizers in place for transfers. She has been positioned by nursing and physical therapy staff throughout the admission with strict adherence to weight-bearing precautions.

Hematologic

Massive transfusion on the day of admission (12 units packed red blood cells, 12 units fresh frozen plasma, 2 units apheresis platelets, 10 units cryoprecipitate; total intraoperative EBL approximately 3,450 mL across the combined damage-control laparotomy and orthopedic external-fixation cases). Acute traumatic coagulopathy (INR 1.6 on arrival, 1.9 intraoperatively) resolved with balanced resuscitation and completion of source control; INR normalized by 10/20/2025. She required a further 2 units of packed red blood cells at the 10/25/2025 definitive fixation case, but no further transfusion thereafter. Hemoglobin at transfer 10.4 g/dL. Iron studies at 12/15/2025 were consistent with anemia of inflammation without absolute iron deficiency; oral iron via PEG was initiated. Post-splenectomy platelet count peaked at 782 K/ μ L on 11/03/2025 and has trended down to 421 K/ μ L at transfer, within the expected post-splenectomy range.

Endocrine / Metabolic

Stress hyperglycemia was managed with an insulin infusion for the first ninety-six hours, then transitioned to a scheduled basal-bolus subcutaneous regimen; she was weaned entirely off insulin by 11/18/2025 with subsequent glucose values in the normal range on tube feeds alone. There is no antecedent history of diabetes. She required stress-dose hydrocortisone briefly (10/18 through 10/21) in the setting of catecholamine-refractory hypotension and was tapered off without difficulty. Electrolytes have been managed per critical-care protocol without persistent derangement. Nutrition support (see Gastrointestinal above) has been guided by the clinical dietitian throughout, with PEG-delivered isotonic formula at 60 mL/hr providing approximately 25 kcal/kg/day and 1.5 g/kg/day of protein at goal.

Skin / Wound

Multiple road-rash abrasions across the torso and extremities were managed with routine wound care; all had epithelialized by mid-November. The left iliac-crest, right mid-tibial, and left anteromedial ankle open-fracture wounds are as described under Musculoskeletal, with the left ankle flap and STSG healing without complication. A stage-2 sacral pressure injury developed during the deeply-sedated phase (identified 10/29/2025) and healed with pressure redistribution, low-air-loss mattress, and routine dressings; the sacral skin was intact at transfer.

Consultative Services

Active during this admission and available for continuity with the rehabilitation team:

- **Trauma Surgery** (Dr. Ramaswamy) — primary admitting service through 11/15/2025.
- **Neurosurgery** — ICP monitor placement and management; serial imaging; signed off 12/18/2025.
- **Orthopedic Trauma Surgery** (Dr. Sundararajan) — staged skeletal management; will follow in outpatient clinic at four weeks post-transfer.
- **Plastic and Reconstructive Surgery** (Dr. Hasegawa) — left ankle rotational flap and STSG; will follow at six weeks.
- **Otolaryngology – Head & Neck Surgery** (Dr. Vaccaro) — tracheostomy placement and decannulation.
- **General Surgery** (Dr. Kirkham) — PEG placement; PEG remains in use.
- **Urology** (Dr. Nordstrom) — bladder-dome repair and follow-up cystography; recommendation for outpatient urodynamics after neurogenic-bladder stabilization.
- **Infectious Disease** — VAP and CAUTI treatment planning.
- **Neurology** — non-operative TBI co-management; anti-epileptic management; signed off 12/18/2025.
- **Physical Medicine & Rehabilitation** (Dr. Vasquez-Ortiz, physiatry) — consulted 11/20/2025; author of the rehabilitation transfer summary and the receiving physiatrist of record at Alder Creek Neuro-Rehabilitation Institute.
- **Speech-Language Pathology** — dysphagia evaluation, modified-barium-swallow study, and communication assessment; recommendations continue in rehabilitation.
- **Physical and Occupational Therapy** — bedside range-of-motion and progressive out-of-bed activity within weight-bearing precautions from 11/10/2025 forward; recommendations continue in rehabilitation.
- **Clinical Nutrition** — PEG regimen management throughout.
- **Psychiatry / Neuropsychology** — consulted 12/05/2025 for baseline behavioral assessment; formal serial neuropsychological testing deferred to the rehabilitation setting given the persistence of post-traumatic amnesia at transfer.
- **Trauma Social Work / Case Management** — surrogate identification, conservatorship coordination, insurance authorization, and transfer coordination throughout.

Decision-Making and Surrogate Consent

Throughout the admission Ms. Reyes lacked the capacity to make or to communicate medical decisions and did not participate in informed consent for any procedure or in any goals-of-care discussion. All operative and procedural consents were obtained either by two-physician emergency documentation of surgical necessity or from her surrogate, Ms. Marisela Reyes (sister), who was engaged from the day of admission by trauma social work. A temporary conservatorship of the person was granted by the probate court on 11/10/2025 with Ms. Marisela Reyes appointed

RIVERGATE REGIONAL TRAUMA CENTER

Reyes, Yolanda

Level I Trauma Center

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temporary conservator; that appointment ratified the earlier surrogate consents on file and served as the source of consent for all subsequent care through the date of transfer. Ms. Reyes has not been assessed as decisional at any point during this admission.

STATUS AT TRANSFER (12/22/2025)

- **Neurologic:** Persistent severe cognitive impairment; post-traumatic amnesia not yet resolved (GOAT 71 on 12/20/2025); GCS 13 (E4 V4 M5); intermittently oriented to self only; requires 24-hour supervision; not decisional.
- **Airway / Pulmonary:** Room air, tracheostomy decannulated 11/28/2025, stoma closing by secondary intention.
- **Cardiovascular:** Normotensive, sinus rhythm 78–92 bpm on propranolol.
- **GI / Nutrition:** PEG-fed at goal; NPO by mouth pending swallow rehabilitation.
- **GU:** Intermittent straight catheterization every six hours; pending neurogenic-bladder work-up.
- **Musculoskeletal:** Non-weight-bearing bilateral lower extremities; healing surgical incisions; left ankle flap/STSG intact.
- **Skin:** Intact; sacral pressure injury healed.
- **Functional:** Dependent for all activities of daily living; bed-to-chair transfers with two-person maximum assistance and mechanical lift; unable to ambulate.
- **Infectious:** No active infection at transfer; all cultures negative.

Condition: Stable for transfer to inpatient neurologic rehabilitation.

Disposition: Transferred 12/22/2025 to Alder Creek Neuro-Rehabilitation Institute, 4501 Recovery Parkway, Rivergate, CA 91766, for continued neurologic rehabilitation, weight-bearing progression, swallow and communication rehabilitation, neurogenic-bladder management, and cognitive rehabilitation. Warm hand-off performed by telephone to the receiving physiatrist (Dr. Helena Vasquez-Ortiz, MD). PEG, wound care, medication reconciliation, and orthopedic weight-bearing precautions communicated in writing and confirmed verbally.

Code Status at Transfer: Full code, per surrogate.

Electronically signed by:

Andrew Steinbach, MD

Attending Intensivist — Surgical / Neurosurgical Critical Care

Rivergate Regional Trauma Center

12/22/2025 14:20